

MSAJCE — Mohamed Sathak A.J. College of Engineering

ECE w/s in Advanced Communication Technology — Complete Data Extract for RAG Model

Source URL: <https://www.msajce-edu.in/ece-act.php> | Sections: 17 sections covered

Scraped: 2026-04-29 | Chunks: 9 | Token Target: 500 | Overlap: 100 | Model: text-embedding-3-small

1. Programme Overview — B.Tech ECE with Specialization in Advanced Communication Technology

Chunk ID: msajce_ece-act_chunk_01 | URL: <https://www.msajce-edu.in/ece-act.php#tab1> | ~420 tokens

Describes the B.Tech ECE Advanced Communication Technology programme, its focus areas, technologies covered, and industry sectors students can enter after graduation.

B.Tech ECE with Specialization in Advanced Communication Technology — MSAJCE

B.Tech – ECE Specialization in Advanced Communication Technology, under the faculty of Electronics and Communication Engineering, has been framed based on NEP guidelines, with a focus upon Multidisciplinary education.

The objective of the program is to develop Advanced Communication Technology based industry competent professionals of global standard capable of accepting challenges and able to solve societal problems at a larger level.

The world getting digitized and the speed of communication becoming an important factor, the evolution of 5G/6G communication systems is imminent. The B.Tech program in ECE Specialization in Advanced Communication Technology focuses on modern communication systems such as:

- Wireless & Optical Communication
- RF & Microwave Communication
- Laser & Light Wave Communication
- Next Generation (5G / 6G) Communication
- Advanced Communication Technologies: Massive Multiple-Input Multiple-Output (MIMO), Beam Forming through Advanced Digital Signal Processing, Device to Device (D2D) communication, Millimeter Wave (mm-Wave) technologies, Communication Networks, Embedded Systems, IoT, Cyber & Network Security, Information Theory & Coding
- Satellite Communication, Navigation systems, Quantum and Molecular Communication

A strong emphasis is placed on developing practical skills and in-depth knowledge required by today's industries.

Industry Sectors / Employers:

Network providers: BSNL, AIRTEL; Manufacturing, Health care, Education, IT sector; Civil Aviation, Defense, Postal and Railway Departments; All India Radio & TV stations; Indian Telecom Sector; National Physics Lab; Bharat Electronics Ltd; ONGC, SAIL, and PSUs; Cisco, Havells, 3M India, Reliance, Bajaj, Bosch, Philips and many more.

Curriculum Link: <https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf>

Student Portal: <http://msajce-edu.in/msajce-student-portal/index.php>

Sample Questions this chunk answers:

- What is B.Tech ECE Advanced Communication Technology at MSAJCE?
- What technologies are covered in the ECE-ACT programme at MSAJCE?
- Which companies hire ECE Advanced Communication Technology graduates from MSAJCE?
- Is the ECE-ACT programme at MSAJCE based on NEP guidelines?
- What is 5G/6G and is it taught at MSAJCE ECE-ACT?
- Does MSAJCE ECE-ACT cover satellite communication?
- What is MIMO and is it part of the MSAJCE ECE-ACT syllabus?
- Can I join BSNL or DRDO after studying ECE-ACT at MSAJCE?

Keywords: B.Tech ECE, Advanced Communication Technology, ACT, 5G, 6G, MIMO, mm-Wave, millimeter wave, D2D, IoT, RF, Microwave, Optical Communication, Satellite Communication, Quantum Communication, Embedded Systems, NEP, multidisciplinary, BSNL, AIRTEL, Bharat Electronics, CISCO, Bosch, network security, information theory

2. Programme Details — Duration, Seats, Eligibility, Vision and Mission

Contains the key programme details for B.Tech ECE-ACT including duration, seat count, eligibility criteria, and the department's vision and mission statements.

B.Tech ECE Advanced Communication Technology — Programme Details

Duration: 4 years (Regular) / 3 years (Lateral Entry)

No. of Semesters: 8 (Regular) / 6 (Lateral Entry)

Intake / No. of Seats: Total — 30 (Government — 15, Management — 15)

Eligibility: 10+2 system of Education. Must have secured a pass in Physics, Chemistry and Mathematics in the qualifying examination.

Curriculum & Syllabus (2021 Regulation): <https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf>

Scope for Higher Studies: M.E. / M.Tech. / M.B.A.

Vision:

To be a centre of excellence for transforming students into proficient Communication Engineers through sustainable practices.

Mission:

M1. Impart core fundamental knowledge and necessary skills in communication systems through innovative teaching and learning methodology.

M2. Inculcate critical thinking, ethics, lifelong learning and creativity needed for industry and society.

M3. Cultivate the students with all-round competencies, for career, higher education and self-employability.

Note: NRI Quota applies — 5% of sanctioned seats (approximately 1–2 seats) are reserved under NRI quota as per standard Anna University / TNEA norms.

B.Tech ECE-ACT Programme Details

Parameter	Details
Programme	B.Tech ECE with Specialization in Advanced Communication Technology
Duration	4 years (Regular) / 3 years (Lateral Entry)
Semesters	8 (Regular) / 6 (Lateral Entry)
Total Seats	30
Government Seats	15
Management Seats	15
Eligibility	10+2 with Physics, Chemistry, Mathematics (pass)
Higher Studies	M.E. / M.Tech. / M.B.A.
Curriculum	2021 Regulation — https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf

Sample Questions this chunk answers:

- How many seats are available in B.Tech ECE-ACT at MSAJCE?
- What is the duration of the ECE Advanced Communication Technology programme at MSAJCE?
- What is the eligibility for B.Tech ECE-ACT at MSAJCE?
- How many government and management quota seats are there in ECE-ACT at MSAJCE?
- Can I do lateral entry into ECE-ACT at MSAJCE?
- What is the vision of the ECE-ACT department at MSAJCE?
- What higher studies can I pursue after B.Tech ECE-ACT at MSAJCE?
- What is the mission of the ECE Advanced Communication Technology department at MSAJCE?

Keywords: ECE-ACT seats, intake, 30 seats, government quota, management quota, lateral entry, eligibility, Physics Chemistry Mathematics, 2021 Regulation, M.E., M.Tech., M.B.A., vision, mission, communication engineer, sustainable, NEP

[Overlap carry-forward] Industry Sectors: BSNL, AIRTEL, Civil Aviation, Defense, Postal and Railway Departments, All India Radio & TV stations, National Physics Lab, Bharat Electronics Ltd, ONGC, SAIL, Cisco, Havells, 3M India, Reliance, Bajaj, Bosch, Philips.

3. Programme Educational Objectives, Programme Specific Outcomes and Programme Outcomes (PEOs, PSOs, POs)

Chunk ID: msajce_ece-act_chunk_03 | URL: <https://www.msajce-edu.in/ece-act.php#tab2> | ~480 tokens

Lists all Programme Educational Objectives (PEOs), Programme Specific Outcomes (PSOs), and the 12 Programme Outcomes (POs) for the B.Tech ECE-ACT programme.

Programme Educational Objectives (PEOs) — B.Tech ECE-ACT, MSAJCE

PEO1. Graduates will be prepared for analysing, designing, developing and testing the software solutions and products with creativity and sustainability.

PEO2. Graduates will be skilled in the use of modern tools for critical problem solving and analyzing industrial and societal requirements.

PEO3. Graduates will be prepared with managerial and leadership skills for career and starting up own firms.

Programme Specific Outcomes (PSOs) — Engineering Graduates will be able to:

PSO1. Develop creative solutions by adapting emerging technologies/tools for real time applications of Industry.

PSO2. Apply the acquired knowledge to develop software solutions and innovative mobile apps for various automation applications.

Programme Outcomes (POs):

PO1. Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to solve complex engineering problems.

PO2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes meeting specified needs with consideration for public health, safety, and cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex Problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools for complex engineering activities.

PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of engineering solutions in societal and environmental contexts, and demonstrate knowledge of sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities, including writing effective reports, design documentation, presentations, and giving clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge of engineering and management principles, manage projects, and work in multidisciplinary environments.

PO12. Life-long Learning: Recognize the need for, and have the ability to engage in independent and life-long learning in the context of technological change.

Sample Questions this chunk answers:

- What are the Programme Educational Objectives of ECE-ACT at MSAJCE?
- What are the Programme Specific Outcomes for ECE-ACT at MSAJCE?
- What are the Programme Outcomes (POs) for B.Tech ECE at MSAJCE?
- Does the ECE-ACT programme at MSAJCE teach ethics and sustainability?
- What skills will I develop in B.Tech ECE-ACT at MSAJCE?

- Does ECE-ACT at MSAJCE prepare students for entrepreneurship?
- What is PO12 for ECE-ACT at MSAJCE?
- Does the ECE-ACT programme focus on project management?

Keywords: PEO, PSO, PO, Programme Educational Objectives, Programme Specific Outcomes, Programme Outcomes, OBE, engineering knowledge, problem analysis, design solutions, modern tools, ethics, sustainability, team work, project management, life-long learning, communication skills, automation, mobile apps

[Overlap carry-forward] B.Tech ECE-ACT: Total Seats 30 (Government 15, Management 15). Duration 4 years / 8 semesters. Eligibility: 10+2 with Physics, Chemistry, Mathematics. Vision: To be a centre of excellence for transforming students into proficient Communication Engineers.

4. Job Profiles, Job Descriptions and Average Annual Salaries for ECE-ACT Graduates

Chunk ID: msajce_ece-act_chunk_04 | URL: <https://www.msajce-edu.in/ece-act.php#tab3> | ~500 tokens

Lists 10 job profiles available to ECE Advanced Communication Technology graduates with their job descriptions and average annual salaries in India.

Job Profiles and Job Descriptions — Electronics and Communication Engineers (ECE-ACT)

- 1 | Design Engineer | Work on research and design of new products — draft and test prototypes, ensure smooth manufacturing process.
- 2 | ASIC Engineer | Find and understand system complexities, devise solutions to enhance system performance.
- 3 | Embedded Engineer | Design and implement software used by embedded devices and systems. Participate in all steps of software development life cycle.
- 4 | Network Support Engineer | Ensure communication systems that organizations and individuals work on run smoothly and without bugs, on both external and internal levels.
- 5 | Software Engineer | Identify needs for new software programs, design and develop them, improve existing ones in terms of UI, speed, and new features. Check hardware and software work properly.
- 6 | Quality Analyst | Actively participate in development and planning of various test and process plans and test scripts for a project.
- 7 | Technician | Participate in first few phases of software development life cycle including prototyping, identifying and eliminating potential threats.
- 8 | Chief Technical Officer | Overlook the technical sector of an organization on a managerial level and work towards achieving growth targets.
- 9 | ECE Engineering Consultant | Act as a bridge between the client and working team to ensure client needs are met and technical solutions comply with industry regulations.
- 10 | Data Analyst | Analyze raw data from different sectors, derive meaningful results (graphs, charts) to help in decision-making and ensure maximum profitable solutions.

Average Annual Salary of ECE-ACT Graduates in India:

- 1 | Design Engineer | 4 – 5 lakh
- 2 | ASIC Engineer | 20 lakh
- 3 | Embedded Engineer | 16 lakh
- 4 | Network Support Engineer | 3 – 4 lakh
- 5 | Software Engineer | 4 – 12 lakh
- 6 | Quality Analyst | 4.5 lakh
- 7 | Technician | 2 – 2.5 lakh
- 8 | Chief Technical Officer | 18 lakh
- 9 | ECE Engineering Consultant | 20 lakh
- 10 | Data Analyst | 5 – 8 lakh

Job Profiles and Average Annual Salary — ECE-ACT Graduates

S.No	Job Profile	Average Annual Salary (Rs.)
1	Design Engineer	4 – 5 lakh

S.No	Job Profile	Average Annual Salary (Rs.)
2	ASIC Engineer	20 lakh
3	Embedded Engineer	16 lakh
4	Network Support Engineer	3 – 4 lakh
5	Software Engineer	4 – 12 lakh
6	Quality Analyst	4.5 lakh
7	Technician	2 – 2.5 lakh
8	Chief Technical Officer	18 lakh
9	ECE Engineering Consultant	20 lakh
10	Data Analyst	5 – 8 lakh

Sample Questions this chunk answers:

- What are the job opportunities after B.Tech ECE-ACT at MSAJCE?
- What is the salary of an ASIC Engineer in India after ECE?
- What does a Design Engineer do after ECE?
- What is the average salary of an ECE graduate in India?
- What is the salary of an Embedded Engineer in India?
- Can I become a Chief Technical Officer after ECE-ACT at MSAJCE?
- What is the salary of a Data Analyst with ECE background in India?
- What is an ECE Engineering Consultant and what is the salary?

Keywords: job profiles, salary, Design Engineer, ASIC Engineer, Embedded Engineer, Network Support Engineer, Software Engineer, Quality Analyst, CTO, Chief Technical Officer, ECE Consultant, Data Analyst, annual salary, lakh, ECE career, placement, employment

[Overlap carry-forward] PO12. Life-long Learning: Recognize the need for and have the ability to engage in independent and life-long learning in the context of technological change. ECE-ACT prepares students with managerial and leadership skills for career and starting up own firms (PEO3).

5. Employment Forecast 2025–2030 — Key Industry Drivers and Sector-Wise Job Growth

Chunk ID: msajce_ece-act_chunk_05 | URL: <https://www.msajce-edu.in/ece-act.php#tab3> | ~510 tokens

Details the employment forecast for Advanced Communication Technology in India from 2025 to 2030, covering key industry drivers, sector-wise growth rates, and government policy impacts.

Advanced Communication Technology Employment Forecast in India 2025–2030

The field is experiencing rapid growth, driven by innovations in 5G/6G, IoT, satellite communications, and wireless technologies.

Key Industry Drivers:

- 5G and 6G Deployment:** India's 5G rollout (and preparations for 6G) will lead to significant hiring in network planning, RF design, and protocol development. Growth in telecom infrastructure, network security, wireless communication, and millimeter-wave (mm-wave) technology.
- Satellite Communication & Remote Sensing:** India's expanding Space Industry, including ISRO, private satellite companies (e.g., OneWeb, Bharti Airtel's satellite ventures), and global satellite internet networks (e.g., Starlink), will generate demand for satellite communication engineers. Key roles in satellite system design, communications payloads, and ground station operations.
- Internet of Things (IoT):** Explosion of connected devices (smart cities, smart homes, industrial IoT) will require skilled professionals in low-power communication protocols (LPWAN), sensor networks, and embedded systems. Focus on IoT communication protocols (LoRa, Zigbee, NB-IoT), 5G IoT applications, and edge computing.
- Wireless Communication & Broadband:** Demand for high-speed internet will create job openings in wireless broadband, Wi-Fi 6/6E, and optical fiber communication. Development of mmWave-based systems, software-defined radio (SDR), and advanced modulation techniques will create numerous roles.

5. Telecommunications R&D: Increased investment in R&D for next-generation communication technologies (6G, optical wireless, quantum communication) will create positions for research engineers and technology specialists.

Sector-Wise Job Growth Projections 2025–2030:

Telecom (5G/6G) | 10–12% per year | RF engineering, Network design, 5G protocol testing

Satellite Communications | 12–15% per year | Satellite payload design, ground station operations

IoT & Embedded Systems | 15–18% per year | IoT communication protocols, sensor networks, edge computing

Broadband & Wireless | 8–10% per year | mmWave systems, optical fiber, Wi-Fi 6/6E technologies

R&D & Next-Gen Comms | 10–14% per year | Quantum communication, optical wireless, 6G technologies

Sector-Wise Job Growth Projections 2025–2030

Sector	Growth Rate (2025–2030)	Key Hiring Areas
Telecom (5G/6G)	10–12% per year	RF engineering, Network design, 5G protocol testing
Satellite Communications	12–15% per year	Satellite payload design, ground station operations
IoT & Embedded Systems	15–18% per year	IoT communication protocols, sensor networks, edge computing
Broadband & Wireless	8–10% per year	mmWave systems, optical fiber, Wi-Fi 6/6E technologies
R&D & Next-Gen Comms	10–14% per year	Quantum communication, optical wireless, 6G technologies

Sample Questions this chunk answers:

- What is the job growth in telecom sector in India by 2030?
- Which sector has the highest job growth for ECE graduates in India?
- Is IoT a growing field for ECE graduates in India?
- What is the growth rate for satellite communication jobs in India?
- Will 5G create more jobs for ECE graduates?
- What are the key hiring areas in wireless broadband in India?
- What is the demand for quantum communication engineers in India?
- What is the future scope of ECE-ACT in India?

Keywords: employment forecast, 2025 2030, 5G 6G, IoT, satellite communication, ISRO, Starlink, OneWeb, Bharti Airtel, LPWAN, LoRa, Zigbee, NB-IoT, Wi-Fi 6, SDR, software defined radio, mmWave, optical fiber, quantum communication, job growth, telecom sector India, smart cities, edge computing

[Overlap carry-forward] ECE-ACT graduate salaries: ASIC Engineer 20 lakh, Embedded Engineer 16 lakh, CTO 18 lakh, ECE Consultant 20 lakh, Data Analyst 5–8 lakh, Software Engineer 4–12 lakh per year in India.

6. Employment by Role, Government Initiatives and Skills/Certifications Demand

Chunk ID: msajce_ece-act_chunk_06 | URL: <https://www.msajce-edu.in/ece-act.php#tab3> | ~490 tokens

Lists employment opportunities by specific roles for 2025–2030, key government policy impacts on hiring, and the certifications and skills most demanded by the industry.

Employment Opportunities by Role (2025–2030) — Advanced Communication Technology

Telecom Engineer (5G/6G) | High | RF design, Antenna design, mmWave, LTE, 5G NR, network optimization

Satellite Communication Engineer | High | Satellite design, payload integration, frequency planning, communication systems

IoT Network Engineer | Very High | Low-power protocols (LoRa, NB-IoT), sensor networks, embedded systems

Wireless Communication Specialist | Moderate to High | Wi-Fi 6, mmWave, optical fiber, SDR, broadband deployment

Research Engineer (6G/Quantum Comms) | High | Quantum computing, 6G technologies, advanced wireless systems

RF System Engineer | High | Signal processing, RF components, microwave engineering

Network Security Engineer | Very High | Network protocols, cybersecurity, 5G security, encryption

Key Government Initiatives & Policy Impacts:

1. National Digital Communications Policy (NDCP 2018): Aimed at positioning India as a global leader in communication technologies and attracting investment in 5G and beyond.
2. Space Policy: Promoting private-sector participation in satellite communications.
3. Smart Cities Mission: Generating demand for IoT, broadband, and wireless communication infrastructure.
4. PLI Scheme (Production Linked Incentive): Boosting manufacturing of telecom equipment and network components in India, leading to job creation.

Skills and Certifications Demand in Industry:

1. 5G & 6G Expertise: Certifications from 3GPP, IEEE, and Cisco.
2. IoT & Embedded Systems: Courses on Arduino, Raspberry Pi, STM32, and LoRaWAN.
3. Satellite Communications: Hands-on experience with Satcom equipment and ground stations.
4. Wireless Protocols: Knowledge of Wi-Fi 6, Bluetooth 5.0/5.1, LoRa, and NB-IoT.
5. Software-Defined Networking (SDN): Familiarity with SDN and Network Function Virtualization (NFV).

Employment Opportunities by Role 2025–2030

Role	Job Growth (2025–2030)	Skills Required
Telecom Engineer (5G/6G)	High	RF design, Antenna design, mmWave, LTE, 5G NR, network optimization
Satellite Communication Engineer	High	Satellite design, payload integration, frequency planning, communication systems
IoT Network Engineer	Very High	Low-power protocols (LoRa, NB-IoT), sensor networks, embedded systems
Wireless Communication Specialist	Moderate to High	Wi-Fi 6, mmWave communication, optical fiber, SDR, broadband deployment
Research Engineer (6G/Quantum)	High	Quantum computing, 6G technologies, advanced wireless systems
RF System Engineer	High	Signal processing, RF components, microwave engineering

Role	Job Growth (2025–2030)	Skills Required
Network Security Engineer	Very High	Network protocols, cybersecurity, 5G security, encryption

Sample Questions this chunk answers:

- What skills do I need to become a Network Security Engineer after ECE-ACT?
- What certifications should I get after B.Tech ECE-ACT at MSAJCE?
- What is the demand for IoT Network Engineers in India?
- How does the PLI scheme create jobs for ECE graduates?
- What is NDCP 2018 and how does it affect ECE job opportunities?
- What skills are needed for a 5G/6G Telecom Engineer?
- Which ECE roles have very high job growth by 2030?
- What are the certifications in demand for ECE-ACT graduates?

Keywords: Telecom Engineer, Satellite Communication Engineer, IoT Network Engineer, Network Security Engineer, RF System Engineer, Wireless Communication Specialist, Research Engineer, 5G NR, LTE, antenna design, 3GPP, IEEE, Cisco, Arduino, Raspberry Pi, STM32, LoRaWAN, Bluetooth 5.0, SDN, NFV, NDCP 2018, PLI scheme, Smart Cities Mission, Space Policy

[Overlap carry-forward] Sector-wise growth 2025–2030: IoT & Embedded Systems 15–18%/year (highest), Satellite Communications 12–15%/year, R&D & Next-Gen Comms 10–14%/year, Telecom (5G/6G) 10–12%/year, Broadband & Wireless 8–10%/year.

7. Faculty Details — ECE w/s in Advanced Communication Technology

Chunk ID: msajce_ece-act_chunk_07 | URL: <https://www.msajce-edu.in/ece-act.php> | ~430 tokens

Lists all faculty members of the ECE-ACT department at MSAJCE with their names, qualifications, designations, and areas of specialization.

Faculty — ECE with Specialization in Advanced Communication Technology, MSAJCE

Dr. NAGASUBRAMANIAN R | M.E., Ph.D. | Head - PG | Computer Science Engineering

Dr. SUBRAMANIAN P | M.E., Ph.D. | Head - UG | Wireless Sensor Networking and Image Processing

Dr. PREMALATHA G | M.E., Ph.D. | Associate Professor | Image Processing and Machine Learning

Mrs. SARANYA V | M.Tech., (Ph.D) | Assistant Professor | IoT and Cloud Computing

Mrs. SASIKALA L | M.E. | Assistant Professor | Computer Science and Engineering

Mrs. KIRUTHIKA S | M.E. | Assistant Professor | Computer Science and Engineering

Mrs. ASRIN MAHMOOTHA A | M.E. | Assistant Professor | Computer Science Engineering

Mr. PRAVEENKUMAR P | M.E. | Assistant Professor | Computer Science Engineering

Mr. RAJAKUMAR B | M.Tech., (Ph.D) | Assistant Professor | IoT, AI, NLP and Image Processing

Mr. RAJASEKAR R | M.E., (Ph.D) | Assistant Professor | IoT, Network Security

Faculty — ECE-ACT Department

Name	Qualification	Designation	Area of Specialization
Dr. NAGASUBRAMANIAN R	M.E., Ph.D.	Head - PG	Computer Science Engineering

Name	Qualification	Designation	Area of Specialization
Dr. SUBRAMANIAN P	M.E., Ph.D.	Head - UG	Wireless Sensor Networking and Image Processing
Dr. PREMALATHA G	M.E., Ph.D.	Associate Professor	Image Processing and Machine Learning
Mrs. SARANYA V	M.Tech., (Ph.D)	Assistant Professor	IoT and Cloud Computing
Mrs. SASIKALA L	M.E.	Assistant Professor	Computer Science and Engineering
Mrs. KIRUTHIKAS	M.E.	Assistant Professor	Computer Science and Engineering
Mrs. ASRINIMATHA A	M.E.	Assistant Professor	Computer Science Engineering
Mr. PRAVEENKUMAR P	M.E.	Assistant Professor	Computer Science Engineering
Mr. RAJAKUMAR B	M.Tech., (Ph.D)	Assistant Professor	IoT, AI, NLP and Image Processing
Mr. RAJASEKAR R	M.E., (Ph.D)	Assistant Professor	IoT, Network Security

Sample Questions this chunk answers:

- Who is the Head of Department for ECE-ACT at MSAJCE?
- Who teaches IoT at the ECE-ACT department of MSAJCE?
- Who are the faculty members of ECE-ACT at MSAJCE?
- Does ECE-ACT at MSAJCE have faculty specializing in machine learning?

- Who is the UG head of ECE-ACT at MSAJCE?
- What is Dr. SUBRAMANIAN P's specialization at MSAJCE?
- Does MSAJCE ECE-ACT have a faculty member for network security?
- Who teaches AI and NLP in ECE-ACT at MSAJCE?

Keywords: Dr. Nagasubramanian, Dr. Subramanian P, Dr. Premalatha G, Saranya V, Sasikala L, Kiruthika S, Asrin Mahmootha, Praveenkumar P, Rajakumar B, Rajasekar R, Head of Department, faculty, ECE-ACT staff, IoT, Cloud Computing, Machine Learning, Image Processing, Network Security, NLP, AI, Wireless Sensor Networking

[Overlap carry-forward] Skills and Certifications Demand: 5G/6G — 3GPP, IEEE, Cisco certifications; IoT — Arduino, Raspberry Pi, STM32, LoRaWAN; Satellite — Satcom equipment; Wireless — Wi-Fi 6, Bluetooth 5.0, LoRa, NB-IoT; SDN and NFV familiarity.

8. Laboratory Facilities — ECE-ACT Department

Chunk ID: msajce_ece-act_chunk_08 | URL: <https://www.msajce-edu.in/ece-act.php> | ~430 tokens

Lists all laboratory facilities available in the ECE-ACT department at MSAJCE, with course links, virtual lab links, and lab record download details.

Laboratory Facilities — ECE-ACT Department, MSAJCE

1 | C Programming Lab | CS8261 - C Programming | Virtual Lab:

<https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-C-Programming.pdf> | Objectives:

<https://www.msajce-edu.in/images/departments/cse/Obj-c.pdf> | Equipment Details:

<https://www.msajce-edu.in/images/departments/cse/Equipments-C-Programming.pdf>

2 | Internet Programming Lab | CS8661 - Internet Programming | Virtual Lab:

<https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-Internet-Programming.pdf> | Objectives:

<https://www.msajce-edu.in/images/departments/cse/Obj-INTERNET.pdf>

3 | Operating Systems Lab | CS8461 - Operating Systems | Virtual Lab:

<https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-OS.pdf> | Objectives:

<https://www.msajce-edu.in/images/departments/cse/Obj-os.pdf>

4 | Object Oriented Analysis and Design Lab | CS8461 - Object Oriented Analysis and Design | Virtual Lab:

<https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-ood.pdf> | Objectives:

<https://www.msajce-edu.in/images/departments/cse/Obj-ood.pdf>

5 | Web Technology Lab | IT8511 - Web Technology | Virtual Lab:

<https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-Web-Technology.pdf> | Objectives:

<https://www.msajce-edu.in/images/departments/cse/Obj-web.pdf>

6 | Networks Lab | CS8581 - Networks Lab | Virtual Lab: <https://www.msajce-edu.in/images/departments/cse/Networks.pdf> |

Objectives: <https://www.msajce-edu.in/images/departments/cse/Obj-Networks.pdf>

All labs provide soft copy lab records for download. Equipment details are available via the department's lab documentation pages. Course syllabi are accessible via the student portal: <http://msajce-edu.in/msajce-student-portal/index.php>

Laboratory Facilities — ECE-ACT

Sl.No	Lab Name	Course Offered	Virtual Link
1	C Programming Lab	CS8261 - C Programming	https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-C-Programming.pdf
2	Internet Programming Lab	CS8661 - Internet Programming	https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-Internet-Programming.pdf

Sl.No	Lab Name	Course Offered	Virtual Link
3	Operating Systems Lab	CS8461 - Operating Systems	https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-OS.pdf
4	Object Oriented Analysis and Design Lab	CS8461 - OOAD	https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-ooad.pdf
5	Web Technology Lab	IT8511 - Web Technology	https://www.msajce-edu.in/images/departments/cse/Virtual-Lab-Web-Technology.pdf
6	Networks Lab	CS8581 - Networks Lab	https://www.msajce-edu.in/images/departments/cse/Networks.pdf

Sample Questions this chunk answers:

- What labs are available in the ECE-ACT department at MSAJCE?
- Does ECE-ACT at MSAJCE have a Networks Lab?
- Where can I find the virtual lab link for C Programming at MSAJCE?
- Does ECE-ACT at MSAJCE have an Operating Systems Lab?
- What course is offered in the Web Technology Lab at MSAJCE?
- Where can I download lab records for ECE-ACT at MSAJCE?
- Does MSAJCE ECE-ACT provide virtual lab access?
- What is the lab code for Networks Lab at MSAJCE ECE-ACT?

Keywords: C Programming Lab, Internet Programming Lab, Operating Systems Lab, OOAD Lab, Web Technology Lab, Networks Lab, CS8261, CS8661, CS8461, IT8511, CS8581, virtual lab, lab record, ICT tools, student portal, lab equipment

[Overlap carry-forward] ECE-ACT Faculty: Dr. Nagasubramanian R (Head PG), Dr. Subramanian P (Head UG, Wireless Sensor Networking), Dr. Premalatha G (Machine Learning), Mr. Rajakumar B (IoT, AI, NLP), Mr. Rajasekar R (IoT, Network Security). 10 faculty members total.

9. Course Materials, Syllabus and Placement Companies

Chunk ID: msajce_ece-act_chunk_09 | URL: <https://www.msajce-edu.in/ece-act.php> | ~520 tokens

Lists all course materials available for ECE-ACT students with subject codes and names, and the companies where ECE-ACT students have been placed off-campus.

Course Materials — ECE-ACT, MSAJCE (Student Portal: <http://msajce-edu.in/msajce-student-portal/index.php>)

Anna University Regulations Available: BE - 2013 Syllabus, BE - 2017 Syllabus, ME - 2013 Syllabus, ME - 2017 Syllabus, BE - 2017 Regulation

GE8151 | Problem Solving and Python Programming

CS8251 | Programming in C

MA8351 | Discrete Mathematics

CS8351 | Digital Principles and System Design

CS8391 | Data Structures

CS8392 | Object Oriented Programming

EC8395 | Communication Engineering

MA8402 | Probability and Queueing Theory

CS8491 | Computer Architecture

CS8492 | Database Management Systems

CS8451 | Design and Analysis of Algorithms

CS8493 | Operating Systems

CS8494 | Software Engineering

MA8551 | Algebra and Number Theory

CS8591 | Computer Networks

EC8691 | Microprocessors and Microcontrollers

CS8501 | Theory of Computation

CS8592 | Object Oriented Analysis and Design

CS8651 | Internet Programming

CS8691 | Artificial Intelligence

CS8601 | Mobile Computing

CS8602 | Compiler Design

CS8603 | Distributed Systems

MG8591 | Principles of Management

CS8792 | Cryptography and Network Security

CS8791 | Cloud Computing

(Plus Open Electives I & II, Professional Electives I–V)

Lesson plans and question banks: accessible via student portal.

CS8592 Lesson Plan: <https://www.msajce-edu.in/academics/cse/lessonplan/CS8592OOADLP.pdf>

CS8651 Lesson Plan: <https://www.msajce-edu.in/academics/cse/lessonplan/CS8651IPLP.pdf>

Placement — Off Campus Companies (ECE-ACT graduates placed at):

Accenture, AT&T, Byju's, Capgemini, Carbonite, CISCO, Cognizant (CTS), Deloitte Consulting, HCL, HP, HTC, IBM, Infosys, L&T Infotech, Oracle, Solverminds, Sutherland, TCS, Tech Mahindra, Wells Fargo, Wipro, Wolters Kluwer ELM, and others.

Placement page: <https://www.msajce-edu.in/placement.php>

Course Materials — Subject Codes and Names

Subject Code	Subject Name
GE8151	Problem Solving and Python Programming
CS8251	Programming in C
MA8351	Discrete Mathematics
CS8351	Digital Principles and System Design
CS8391	Data Structures
CS8392	Object Oriented Programming
EC8395	Communication Engineering
MA8402	Probability and Queueing Theory
CS8491	Computer Architecture
CS8492	Database Management Systems
CS8451	Design and Analysis of Algorithms
CS8493	Operating Systems
CS8494	Software Engineering

Subject Code	Subject Name
MA8551	Algebra and Number Theory
CS8591	Computer Networks
EC8691	Microprocessors and Microcontrollers
CS8501	Theory of Computation
CS8592	Object Oriented Analysis and Design
CS8651	Internet Programming
CS8691	Artificial Intelligence
CS8601	Mobile Computing
CS8602	Compiler Design
CS8603	Distributed Systems
MG8591	Principles of Management
CS8792	Cryptography and Network Security
CS8791	Cloud Computing

Sample Questions this chunk answers:

- What subjects are taught in B.Tech ECE-ACT at MSAJCE?
- What is the subject code for Communication Engineering at MSAJCE ECE-ACT?
- Does ECE-ACT at MSAJCE cover Artificial Intelligence?
- Which companies have recruited ECE-ACT students from MSAJCE?
- Is Cryptography and Network Security taught in ECE-ACT at MSAJCE?
- Where can I access course materials for ECE-ACT at MSAJCE?
- Does MSAJCE ECE-ACT have Cloud Computing in its syllabus?
- Has TCS or Infosys recruited from MSAJCE ECE-ACT?

Keywords: GE8151, CS8251, CS8391, EC8395, Communication Engineering, CS8591, Computer Networks, CS8691, Artificial Intelligence, CS8791, Cloud Computing, CS8792, Cryptography, EC8691, Microprocessors, CS8601, Mobile Computing, TCS, Infosys, Wipro, HCL, IBM, Accenture, Capgemini, CISCO, Tech Mahindra, Deloitte, Oracle, Wells Fargo, lesson plan, student portal, Anna University, 2017 regulation
 [Overlap carry-forward] Lab facilities include: C Programming Lab (CS8261), Internet Programming Lab (CS8661), Operating Systems Lab, OOAD Lab, Web Technology Lab (IT8511), Networks Lab (CS8581). Virtual lab links available. Student portal: <http://msajce-edu.in/msajce-student-portal/index.php>

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